5 A microwave oven, shown in Fig 5.1, consists of a reflective stainless steel casing, microwave source which is placed on one of the walls of the casing, and also a turntable at the bottom that rotates. The source produces microwave of frequency 2.45 GHz within the oven which is reflected within the casing forming a stationary wave.

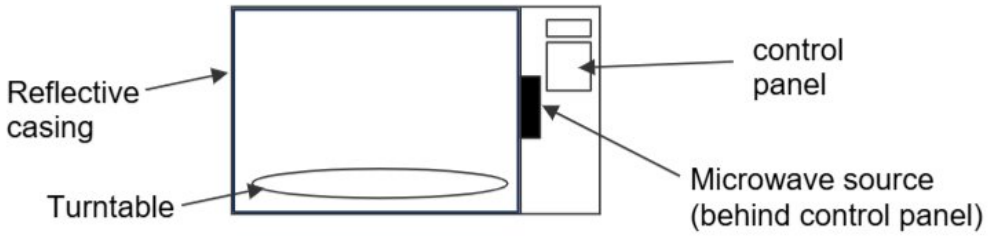


Fig. 5.1

(a) Describe three features of stationary wave.

1 .....

.....

2 .....

.....

3 .....

..... [3]

(b) Explain how stationary waves are formed inside the microwave oven.

.....

.....

.....

..... [2]

(c) A tray of mini marshmallow is arranged in a single layer, as shown in Fig. 5.2. The tray is placed in the microwave oven with the turntable removed. The oven is switched on to cook the marshmallow.



Fig. 5.2

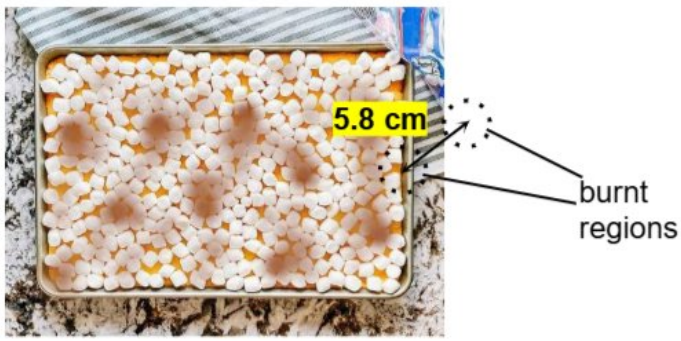


Fig. 5.3

Fig. 5.3 shows the tray of marshmallow taken out from the oven after cooking. It is observed that some regions are burnt more than the other regions. The average length between two adjacent burnt regions is 5.8 cm.

Determine the speed of microwave.

speed = ... m s^{-1} [2]

(d) The door of the microwave oven has a sheet of stainless steel with holes, with diameter about 2 mm, behind the glass, as shown in Fig. 5.4.



Fig. 5.4

Suggest a reason why microwave in the oven cannot pass through the glass while visible light in the oven can pass through.

.....

..... [1]

[Total 8]